EPIB 664 Midterm

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1.

For the following hypothetical scenarios, please assess if the missing data are likely to be MCAR, MAR, and MNAR, and provide your justifications.

(1)

Suppose a survey is held for college students to ask their SAT scores. Since it is a sensitive topic, all these students decided to flip a coin, if it is head, they tell their SAT score, and if it is tail, they refuse to answer. What is the missing data mechanism for the "refused to answer"? (5pt)

Answer

This mechanism would be missing *completely* at random because the students are not deciding whether to report their scores based on the scores themselves, or due to any other variables (this is a univariate problem). That is, the missingness depends only upon an independent "trial" with probability 1/2. This is similar to the first part of Homework 1 (Example 1.2 from lecture), where the missingness of Y under the MCAR mechanism are simple binomial trials - success (1) meaning we denote the variable as "missing", failure (0) meaning we denote the value as observed. In this example, each SAT score could be viewed as a Bernoulli trial with probability one half - which has nothing to do with the *actual values* the students scored. Mathematically (and notationally), this probability one half is reflected in the missingness parameter $\phi = 0.5$, a constant.

(2)

In order to select the "best" students for the AMC 12 math competition, the teacher first asked all the students in the class to take a preliminary math exam. Only students who are higher than 80 points in this preliminary exam are eligible for the AMC 12 math competition. Suppose we know everybody's preliminary exam score, then for the students who do not have AMC 12 math competition score, what is the missingness mechanism of these missing data? (5pt)

Answer

This mechanism is missing at random (MAR), as the missingness depends on the "explanatory" variable, in this case the preliminary exam score. If we know each of the preliminary exam scores, we know for certain which students have an NA in the competition because of that. That is, if we see a student scored a 67 on the preliminary exam, they would have an NA for the competition because they were not eligible for it. In this example, the missingness is dependent upon both the preliminary exam score *and* parameter ϕ, which governs whether or not the students participated in the competition (regardless of whether they qualified).

(3)

Suppose a clinical trial assess the efficacy of a new drug on relieving the chronic pain for late-stage cancer patients. Some patients drop out in the middle of the trial because they don't feel the medication would mitigate their pain. What is the missingness mechanism for the pain level data for these drop-outs? (5pt)

Answer

This mechanism is likely missing not at random (MNAR) because participants with certain levels of pain may be the ones dropping out. A way of looking at this would be to say that the "response" variable (pain level) is missing as a function of itself. In the notation, the missingness of the variable depends on observed values, missing values, and the missingness parameter ϕ. This scenario fits these criteria because a patient with high sustained pain levels (i.e. observed values) may feel that the medication has not been effective, and thus drop out of the study, resulting in missing values. Patients with a relatively extreme (high) level of pain may be less likely to stay in the study because they feel nothing can help. Thus, the data could be systematically missing higher values for pain level, biasing estimates downward, and reflecting an MNAR mechanism.

2.

An investigator would like to research the relationship between people's age and their income. However, some subjects' income values are missing in the dataset. A student proposed to use the mean of the income values from the observed values to impute the missing values. Suppose the association between age and income is positive, that is, older people are likely to earn more than younger people.

(1)

Do you think when using imputed values (the mean number), the association is likely to be increasing, decreasing, or remain the same? State your justifications (5pt)

Answer

The relationship would be attenuated, i.e. its strength would decrease. This is because mean imputation dramatically reduces the variance of the variable that is being imputed (since many instances of the exact same value are present). The method of mean imputation disregards any possible effect of the explanatory variable on the response. That is, we may have a situation where someone aged 18 is imputed the same (mean) income value as someone aged 45. While this is entirely possible from a logistical standpoint, it completely ignores the relationship where older people are more likely to have higher incomes. This can also be visualized with a regression line. The imputed mean points would stand out quite clearly on the scatter plot between age and income as they would be a perfectly straight line across the screen (assuming the missingness is somewhat uniformly distributed across the explanatory variable, attenuation holds either way). A straight horizontal line of course has a slope of 0, as it is a constant, thus the inclusion of several points along this line brings the slope estimate for age closer to 0, decreasing (attenuating) the relationship.

(2)

Another student thinks using the mean value to impute the missing income is not a good idea because the income data tend to be skewed. So he/she proposed to use the median of the observed income values to impute. Same as above, do you think using imputed values, the association is likely to be increasing, decreasing, or remain the same? (5pt)

Answer

This would have a similar effect, as the imputation of the exact same value for *all* missing observations necessarily reduces the variance of the variable. A straight line of points would still result, albeit in a different place if the income data is in fact skewed. In any case, the addition of points in a straight line is always expected to attenuate (or decrease) the relationship between the variables at hand. If anything, median imputation would have a much greater effect on the intercept estimates than the slope estimates, compared to mean imputation. That is, slope estimates will still be attenuated, but if the data is highly skewed to the right, the regression intercept estimate may be lower because there are more "small" values in the data after imputation. The inverse would also be expected - left-skewed data may lead to an larger estimate for the intercept as larger data values are "favored" in the imputation. In all, it is most probable that other methods (i.e. not mean *or* median imputation) would be more favorable for this problem, mainly because they reduce the variance of the explanatory variable.

3.

Suppose you are working with an investigator who is interested in using dgestat.txt (in Lecture 3 folder) to analyze gestational age and related variables. More information about the meaning of these variables can be found in Table 4.3 of my book (page 79). You can also find some background information of this dataset from page 76 to 79 (Example 4.2).
This investigator noticed some of the gestational age variable (dgestat) has missing values. He/she is interested in the following analyses.

(1)

How much missing data does this variable (dgestat) have? Do other variables in the dataset also have missing values? (5pt)

```r
dge_data <- read.table(file="C:/Users/jacma/OneDrive/school work/UMD/EPIB 664 Missing Data/dgestat.txt"
n_missing <- apply(dge_data,2,function(col){sum(is.na(col))})
rbind(Missing_Count = n_missing,
      Missing_Rate = n_missing/nrow(dge_data))
```

```
##                    dgestat dbirwt dmage dfage fmaps wtgain nprevis cigar drink
## Missing_Count 4142.0000000      0     0     0     0      0       0     0     0
## Missing_Rate     0.1691026      0     0     0     0      0       0     0     0
```

```r
# There are 4142 missing values for dgestat, representing about 17% of the rows.
# None of the other variables have *any* missing values
rm(n_missing)
```

(2)

What is the likely missing data mechanism of the variable dgestat? Since the investigator is not a statistician, please interpret your findings to him/her the implication of the missing data mechanism that you identified rather than just tell him/her the terminology. (10 pt)

Propensity Model Approach

```
library(tidyverse)
dge_data$Observed <- ifelse(dge_data$dgestat %>% is.na(),0,1)
propensity_model <- glm(Observed~dbirwt+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
    family=binomial(link="logit"), data=dge_data)
summary(propensity_model)
```

```
##
## Call:
## glm(formula = Observed ~ dbirwt + dmage + dfage + fmaps + wtgain +
##     nprevis + cigar + drink, family = binomial(link = "logit"),
##     data = dge_data)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.1782   0.5272   0.5804   0.6307   1.0906
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.1603173  0.2354276  -0.681    0.496
## dbirwt       0.0001602  0.0000303   5.286 1.25e-07 ***
## dmage        0.0287338  0.0044577   6.446 1.15e-10 ***
## dfage       -0.0003986  0.0038895  -0.102    0.918
## fmaps        0.0394091  0.0245082   1.608    0.108
## wtgain       0.0012864  0.0012931   0.995    0.320
## nprevis      0.0065305  0.0047074   1.387    0.165
## cigar       -0.0236767  0.0040134  -5.899 3.65e-09 ***
## drink       -0.0590985  0.0340954  -1.733    0.083 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 22263  on 24493  degrees of freedom
## Residual deviance: 22060  on 24485  degrees of freedom
## AIC: 22078
##
## Number of Fisher Scoring iterations: 4
```

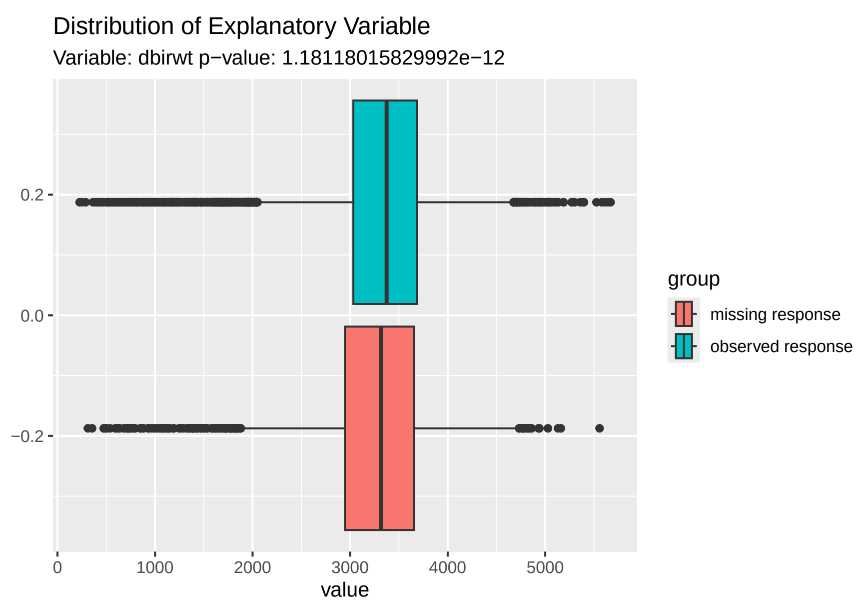
The results tell us that dbirwt, dmage, and cigar are extremely significant in explaining the missingness indicator, which suggests MCAR is unlikely. This means that the distributions of these variables across their response and non-response groups are significantly different. It also implies that these three variables are *most* influential in the missingness of dgestat. It should be noted that this approach does not account for any potential multicollinearity. Note also that the parameter estimates for dbirwt and dmage are positive, suggesting that higher values of these variables increase the likelihood of dgestat being observed. On the other hand, the parameter estimate for cigar is negative, meaning higher values decrease the likelihood of dgestat being observed. In other words, specific values of the covariates probably have a significant effect on whether the dgestat value is available (the last three sentences could be the explanation to the investigator).

t Test Approach

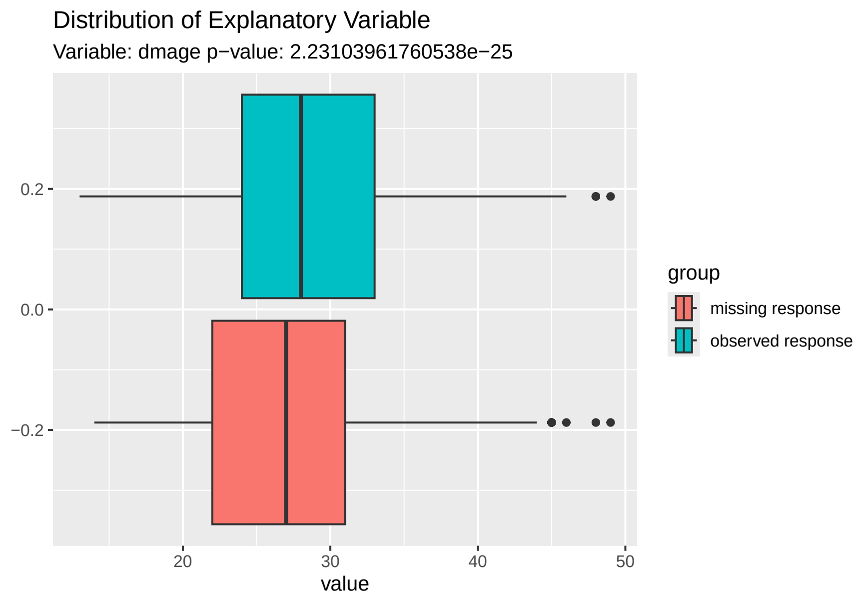
```r
library(reshape2)
mcar_test <- function(missing_data,observed_data,indices){
  # missing and observed data subsets, indices for which columns (covariates)
  # to analyze the difference between
  results <- data.frame()
  missing_longer <- melt(missing_data) # for boxplots
  observed_longer <- melt(observed_data)
  var_names <- missing_longer$variable %>% unique()
  var_names <- var_names[indices]

  pvals <- rep(NA,length(indices))
  missing_means <- pvals # just to get NA vectors of same length
  observed_means <- pvals

  histograms <- list()
  for (i in 1:length(indices)){ # test each column (variable) specified
    column <- indices[i]
    test <- t.test(missing_set[,column],observed_set[,column])
    pvals[i] <- test$p.value # save all test results
    missing_means[i] <- test$estimate[1] %>% round(4)
    observed_means[i] <- test$estimate[2] %>% round(4)

    missing_group <- subset(missing_longer,missing_longer$variable==var_names[i])
    missing_group <- cbind(missing_group,group=rep("missing response",
                                                  nrow(missing_group)))
    observed_group <- subset(observed_longer,observed_longer$variable==var_names[i])
    observed_group <- cbind(observed_group,group=rep("observed response",
                                                    nrow(observed_group)))

    all_data <- rbind(missing_group,observed_group)
    plot <- ggplot(data=all_data) +
      geom_boxplot(mapping=aes(x=value,fill=group)) +
      ggtitle("Distribution of Explanatory Variable",
              paste("Variable:",var_names[i],"p-value:",pvals[i],sep=" "))
    histograms[[i]] <- plot

  }
  results <- data.frame(Variable=var_names,
                   pval=pvals,
                   Missing_Group_Mean = missing_means,
                   Observed_Group_Mean = observed_means)
  return(list(Results=results,Histograms = histograms))
}
missing_set <- subset(dge_data,dge_data$Observed==0)
```

```
observed_set <- subset(dge_data,dge_data$Observed==1)
mcar_test(missing_set,observed_set,2:9)
```

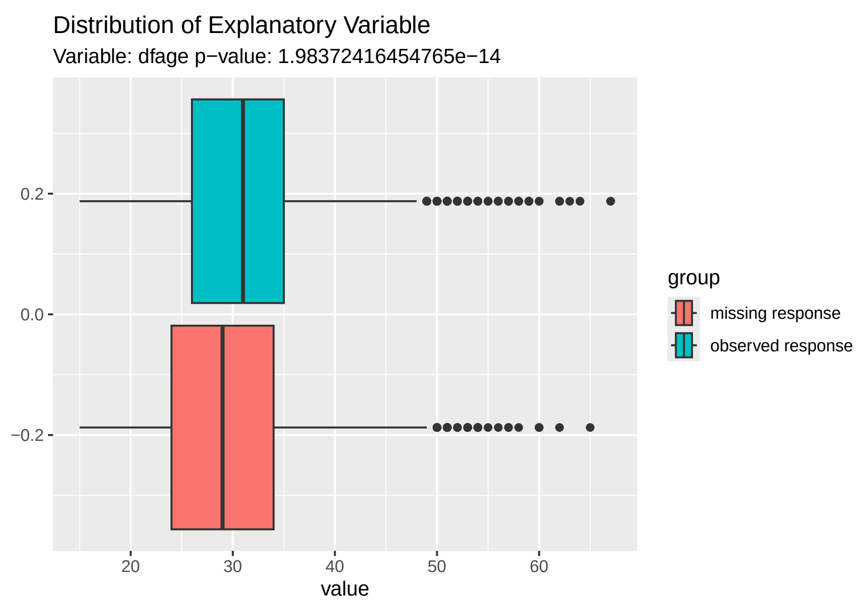
```
## $Results
##   Variable         pval Missing_Group_Mean Observed_Group_Mean
## 1   dbirwt 1.181180e-12          3262.9440           3337.7064
## 2    dmage 2.231040e-25            27.1707             28.2508
## 3    dfage 1.983724e-14            29.8467             30.7547
## 4    fmaps 5.166629e-03             8.8998              8.9347
## 5   wtgain 1.569323e-01            30.7926             31.1313
## 6  nprevis 2.938263e-03            11.5963             11.7931
## 7    cigar 7.390338e-11             1.4073              0.9141
## 8    drink 1.647108e-01             0.0314              0.0161
##
## $Histograms
## $Histograms[[1]]
```



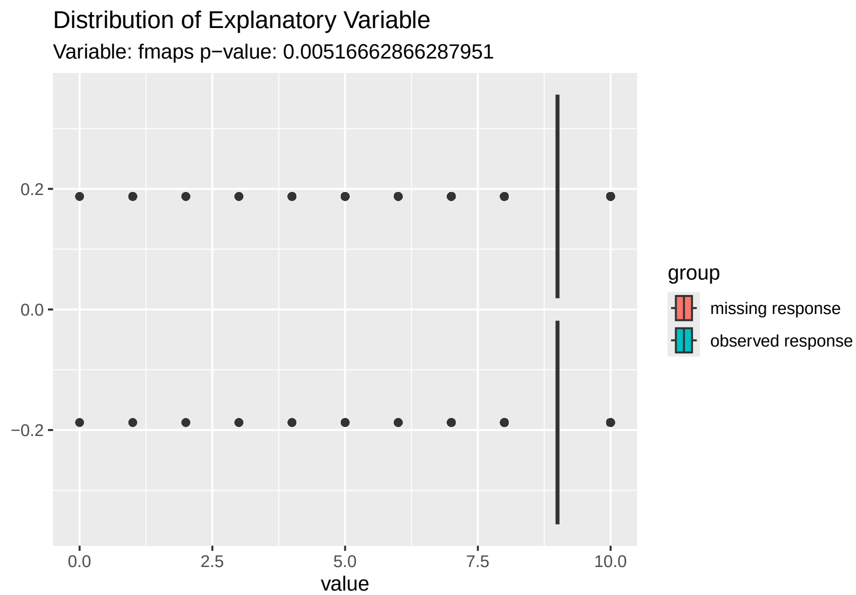

```
##
## $Histograms[[2]]
```



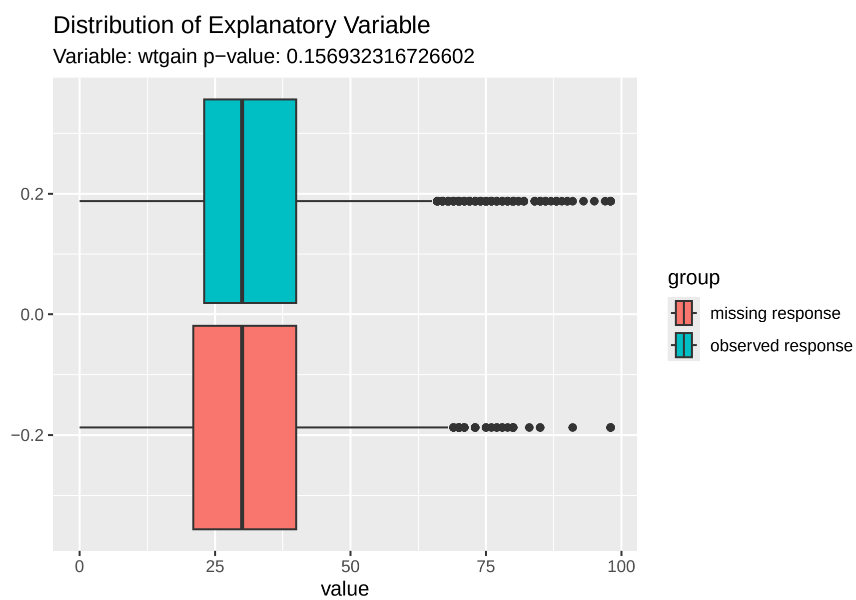

```
##
## $Histograms[[3]]
```



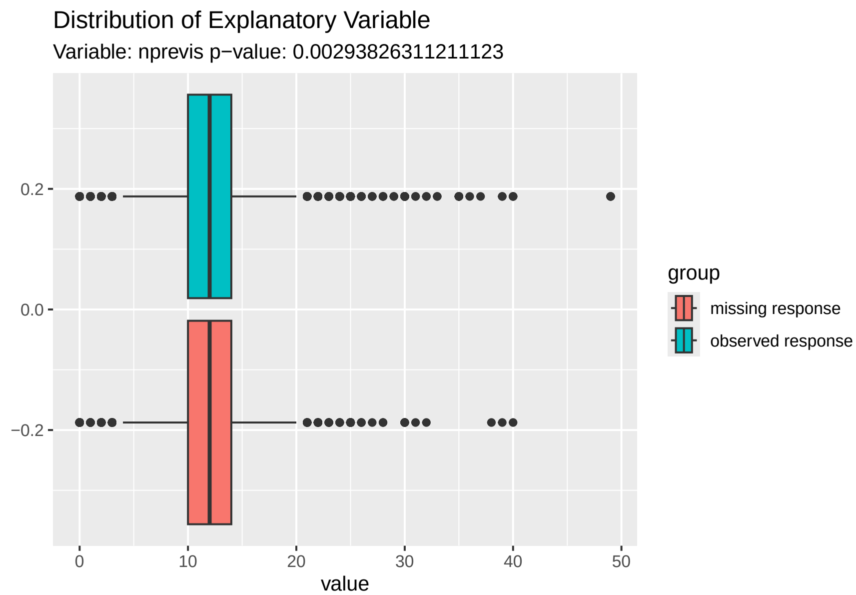

```
##
## $Histograms[[4]]
```



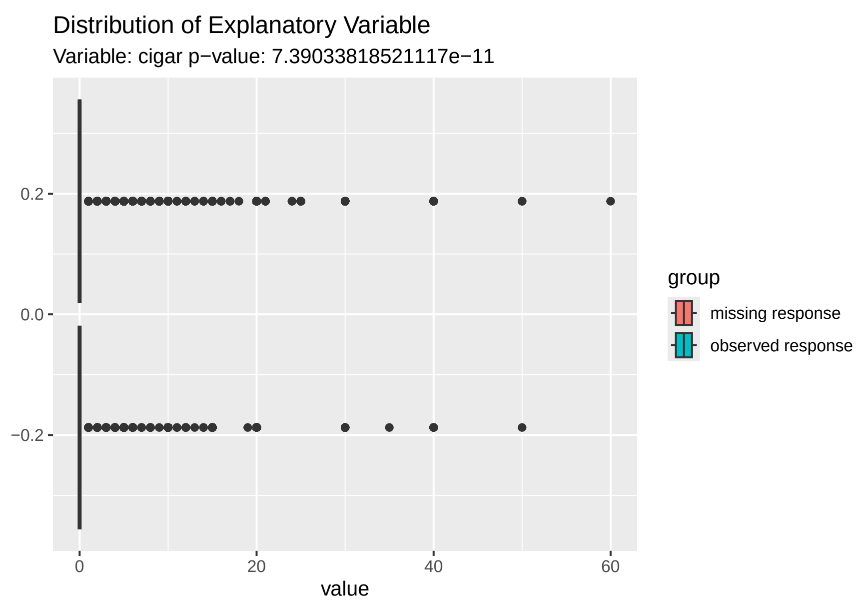

```
##
## $Histograms[[5]]
```



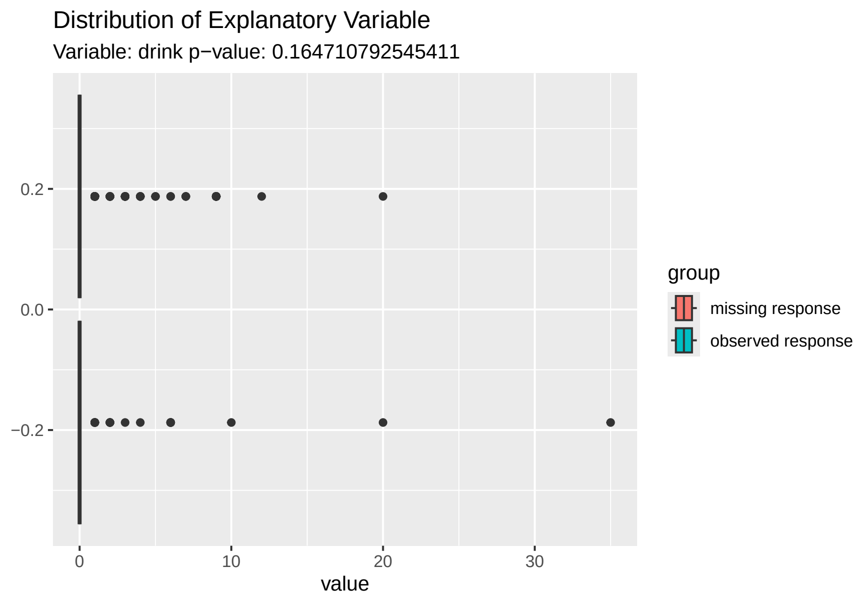

```
##
## $Histograms[[6]]
```




```
##
## $Histograms[[7]]
```




```
##
## $Histograms[[8]]
```




```
# (I probably went a bit overboard with this, but the code could be useful with
# other data sets too)
```

We see from the results table that several of the variables have significantly different means when their corresponding dgestat value is missing vs observed. Moreover, only wtgain and drink are not significantly different at alpha=0.05, while dmage, dfage, dbirwt, and cigar are the "most" significant (in that order). The box plots show the differing dispersion between the variables.
While it is not incredibly easy to distinguish between the groups, we can see that most of the variables have disparate quartiles. We can also see that there are a LOT of outliers for each variable in the data set, and on either end of their respective ranges. This could be another challenge for analysis, and more digging would be required to determine whether outliers have a major effect on missingness.
In layman's terms, the results from these t tests suggest that the values of the aforementioned variables have an effect on whether or not dgestat is observed. So, if we determine significant relationships between dgestat and any of the covariates, said relationship may be altered by the fact certain values may be missing. For example, if we know age has a positive correlation with dgestat, higher values of the dmage variable with missing dgestat could be "hiding" a high value of dgestat. Thus, regression (and other analytical estimates) may not be as accurate as possible due to missingness.
The overall conclusion from the above results is that the missingness mechanism is most likely *not* missing completely at random (MCAR), and is either missing at random (MAR) or missing not at random (MNAR). This is because the significant difference between means for observed and missing responses suggests the value of the covariate likely influences the chance that dgestat is missing. However, distinguishing between MAR and MNAR with data alone is not possible.

(3)

The average of the gestational age (dgestat). The unit of this variable is in weeks (10 pt)

```
### Complete-Case Analysis - 38.67 weeks
base_mean <- dge_data$dgestat %>% mean(na.rm=T)

### Mean Imputation - 38.67 weeks
```

```r
dge_data$mean_impute <- ifelse(dge_data$dgestat %>% is.na(),
                               base_mean,
                               dge_data$dgestat)
# same as above (of course), but variable would have much smaller variance

### Regression Imputation - 38.64 weeks
reg_coef <- lm(dgestat~dbirwt+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
               data=dge_data)$coefficients # fit model, extract coefficients
fitted <- as.matrix(dge_data[,2:9]) %*% reg_coef[2:9] + reg_coef[1] # predictions
dge_data$reg_impute <- ifelse(dge_data$dgestat %>% is.na(),
                              fitted,
                              dge_data$dgestat)
# it should be noted that this method does not account for the random noise
# affecting dgestat (which is assumed to exist by the linear model)

### Nonresponse Weighting - 38.64 weeks
response_probability <- predict(propensity_model, type='response')
response_weights <- 1 / response_probability
dge_data$Weight <- response_weights
dge_data$Weight[is.na(dge_data$dgestat)] <- NA # set unobserved values to NA weight
weight_sum <- sum(dge_data$Weight,na.rm=T) # diagnostic is correct, sum(weights) = ~N
weight_mean <- weighted.mean(dge_data$dgestat,dge_data$Weight, na.rm=T)
full_data <- na.omit(dge_data)
weight_var <- sum(full_data$Weight*(full_data$dgestat-weight_mean)^2) / weight_sum
# *** the formula from the lecture slides was giving a very strange (tiny) value,
# so I used the above formula from online research, result seems to make sense

### EM Algorithm - 38.64 weeks
library(norm)
s <- dge_data[,1:9] %>% as.matrix() %>%
  prelim.norm()
estimates <- em.norm(s) # em algorithm, multivariate normal
```

```
## Iterations of EM:
## 1...2...3...4...5...6...7...
```

```r
em_results <- getparam.norm(s,estimates,corr=T)
# These results also show us that there is not much correlation between any
# pairs of the explanatory variables. The largest correlation estimate is 0.233,
# between dbirwt and fmaps. The only notable correlation between dgestat and the
# covariates is 0.613 with dbirwt.

data.frame(Complete_Case=rbind(Mean=base_mean, # Complete-Case results
                               Var=var(dge_data$dgestat,na.rm=T)),
           Mean_Imputation=rbind(Mean=dge_data$mean_impute %>% mean(), # Mean Impute
                                 Var=dge_data$mean_impute %>% var()),
           Regression_Imputation=rbind(Mean=dge_data$reg_impute %>% mean(), # Regr Impute
                                       Var=dge_data$reg_impute %>% var()),
           Nonresponse_Weighting=rbind(Mean=weighted.mean(dge_data$dgestat, # Weighting
                                                          dge_data$Weight, na.rm=T),
                                       Var=weight_var),
           EM_Algorithm=rbind(Mean=em_results$mu[1],
                              Var=em_results$sdv[1]^2))
```

```
##      Complete_Case Mean_Imputation Regression_Imputation Nonresponse_Weighting
## Mean     38.667797       38.667797             38.644798             38.642945
## Var       4.330056        3.597802              3.969036              4.532498
##      EM_Algorithm
## Mean    38.644807
## Var      4.401594
```

```r
rm(em_results,fitted,propensity_model,s,base_mean,estimates,reg_coef,
   response_probability,response_weights,weight_mean,weight_sum,weight_var)
```

The overall conclusion from these results is that the missing data likely does not have a great effect on analyzing the mean of dgestat. This is because the mean estimate is extremely similar (if not the exact same) across all the methods included, and the variance is only notably reduced if sub-optimal approaches (i.e. mean or regression imputation) are utilized. The table above suggests pretty strongly that the mean of dgestat is around 38.65 weeks. Also note that the mean imputation result is the exact same as the complete case result, but may appear differently due to rounding.

(4)

The regression relationship using gestational age (dgestat) as the outcome variable, and the predictors include the baby's birthweight (dbirwt) and other variables in the dataset. Focus on the coefficient for dbirwt. (20 pt)

```r
# **using imputed values from before saved in dge_data object**
cc_model <- lm(dgestat~dbirwt+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
               data=dge_data) %>% summary()

mean_model <- lm(mean_impute~dbirwt+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
                 data=dge_data) %>% summary()

reg_model <- lm(reg_impute~dbirwt+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
                data=dge_data) %>% summary()


library(survey)
WT_design <- svydesign(id=~1, weights=~Weight, data=full_data[,-c(10,11,12)])
# Apply the survey regression of y on x
weight_model <- svyglm(dgestat~dbirwt+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
                       WT_design,
                       data=full_data[,-c(10,11,12)]) %>% summary()

# extract and format results
data.frame(Complete_Case=rbind(dbirwt_coef=cc_model$coefficients[2],
                               dbirwt_tval=cc_model$coefficients[2,3],
                               dbirwt_pval=cc_model$coefficients[2,4],
                               R2=cc_model$adj.r.squared),
           Mean_Imputation=rbind(dbirwt_coef=mean_model$coefficients[2],
                                 dbirwt_tval=mean_model$coefficients[2,3],
                                 dbirwt_pval=mean_model$coefficients[2,4],
                                 R2=mean_model$adj.r.squared),
           Regr_Imputation=rbind(dbirwt_coef=reg_model$coefficients[2],
```

```r
                                dbirwt_tval=reg_model$coefficients[2,3],
                                dbirwt_pval=reg_model$coefficients[2,4],
                                R2=reg_model$adj.r.squared),
           Weighted_Model=rbind(dbirwt_coef=weight_model$coefficients[2],
                                dbirwt_tval=weight_model$coefficients[2,3],
                                dbirwt_pval=weight_model$coefficients[2,4],
                                R2=NA))
```

```
##              Complete_Case Mean_Imputation Regr_Imputation Weighted_Model
## dbirwt_coef   2.085817e-03     0.001684058    2.085817e-03     0.00213714
## dbirwt_tval   1.024036e+02    94.324215604    1.246910e+02    60.40249002
## dbirwt_pval   0.000000e+00     0.000000000    0.000000e+00     0.00000000
## R2            4.087069e-01     0.326465861    4.640476e-01             NA
```

```r
data.frame(C=cc_model$coefficients[,1:2],
      M=mean_model$coefficients[,1:2],
      R=reg_model$coefficients[,1:2],
      W=weight_model$coefficients[,1:2])
```

```
##               C.Estimate C.Std..Error   M.Estimate M.Std..Error   R.Estimate
## (Intercept) 27.617795836 0.1707402687 30.091335601 1.476646e-01 27.617795836
## dbirwt       0.002085817 0.0000203686  0.001684058 1.785394e-05  0.002085817
## dmage       -0.038724361 0.0029115561 -0.031206111 2.570679e-03 -0.038724361
## dfage        0.005858966 0.0025626320  0.004588945 2.258753e-03  0.005858966
## fmaps        0.529944768 0.0176551192  0.392593129 1.530318e-02  0.529944768
## wtgain      -0.005416237 0.0008610772 -0.004188311 7.547118e-04 -0.005416237
## nprevis      0.035809580 0.0031351347  0.027511651 2.738855e-03  0.035809580
## cigar        0.014715892 0.0031926380  0.013547075 2.681146e-03  0.014715892
## drink       -0.006662522 0.0347228253  0.001657183 2.430041e-02 -0.006662522
##             R.Std..Error   W.Estimate W.Std..Error
## (Intercept) 1.383514e-01 27.247253555 3.746236e-01
## dbirwt      1.672789e-05  0.002137140 3.538166e-05
## dmage       2.408546e-03 -0.038914767 3.024046e-03
## dfage       2.116294e-03  0.005557067 2.682036e-03
## fmaps       1.433801e-02  0.551729674 3.717399e-02
## wtgain      7.071121e-04 -0.005399476 8.910958e-04
## nprevis     2.566115e-03  0.037290551 3.738220e-03
## cigar       2.512046e-03  0.014905192 3.581047e-03
## drink       2.276778e-02 -0.008866074 2.911190e-02
```

```r
# NOTE: The regression models used *all* of the covariates in the data, the R^2
# values reflect that.
# NOTE: C stands for "Complete-Case", M for "Mean Imputation", R for "Regression
# Imputation", and W for "Weighted Model" for the second table

rm(cc_model,mean_model,reg_model,weight_model)
```

The relationship between dbirwt and dgestat is consistently significant across each method. This is shown by the extremely large test statistic and resultant tiny p-values (which display as zero as they are incredibly small). This is strong evidence that **higher values for dbirwt contribute to higher values of dgestat**, owing to the positive parameter estimates in all cases. The small values of the coefficients themselves are due to dbirwt being on a much larger scale than dgestat - thousands compared to tens.

It is also notable that the mean imputation has by far the worst adjusted R^2 value, which is not surprising as the method dramatically reduces the variance of the dependent variable and adds thousands of points along a straight line that is NOT the regression line. Hence, we have a much poorer model fit. Also resulting from this is the attenuation of parameter estimates - as seen for our variable of interest in the first table and for all the others in the second. This is because the aforementioned straight line pulls the slopes of the regression line (and thus the line itself) closer to 0, decreasing the effect of any of the covariates.
Regression imputation having the highest R^2 is also intuitive because it adds thousands of points that fit the line perfectly. The regression imputation also has the exact same estimates as the complete case analysis for the same reason. Another side effect of this "perfect fit adding" is the reduction of standard error estimates, also seen in the table. A more appropriate and reliable way to do this method would be to add in the random fluctuations on dgestat as assumed by the model, though this would also require assuming the variance value of said fluctuations.
The weighted propensity model also had very similar results to the other approaches, albeit with a smaller t value for dbirwt. That said, the value was still *extremely* large suggesting dbirwt's relationship with dgestat remains very strong.
These results lead me to believe that the complete case analysis is sufficient for this problem, because the results of the imputation are very similar *and* come with other caveats. However, regression imputation with the inclusion of random fluctuations on the predicted response would be more statistically reliable as it matches the model assumptions. It largely comes down to whether the analyst prefers to ignore the missing values or not, and since this data has a very high number of observations regardless, the omission of missing data does not seem entirely unfounded. That, and the fact the analytical results are very robust across methods, leads me to this conclusion.
A counterargument could be if the researcher is interested in a high level of precision with the relationship (likely in a prediction setting), in which case the improved regression imputation and/or the propensity model may be more appropriate. If the researcher is only interested in generic insight and the direction of the relationship, any method is sufficient, but complete case is the simplest.

(5)

The regression relationship using the baby's birthweight (dbirwt) as the outcome variable, the gestational age (dgestat) and other variables are predictors. Focus on the coefficient for dgestat. (20 pt)

```r
# **using imputed values from before saved in dge_data object**
cc_model2 <- lm(dbirwt~dgestat+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
               data=dge_data) %>% summary()

mean_model2 <- lm(dbirwt~mean_impute+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
                 data=dge_data) %>% summary()

reg_model2 <- lm(dbirwt~reg_impute+dbirwt+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
                data=dge_data) %>% summary()

weight_model2 <- svyglm(dbirwt~dgestat+dmage+dfage+fmaps+wtgain+nprevis+cigar+drink,
                       WT_design,
                       data=full_data[,-c(10,11,12)]) %>% summary()

# extract and format results
data.frame(Complete_Case=rbind(dbirwt_coef=cc_model2$coefficients[2],
                               dbirwt_tval=cc_model2$coefficients[2,3],
                               dbirwt_pval=cc_model2$coefficients[2,4],
                               R2=cc_model2$adj.r.squared),
           Mean_Imputation=rbind(dbirwt_coef=mean_model2$coefficients[2],
```

```
                                   dbirwt_tval=mean_model2$coefficients[2,3],
                                   dbirwt_pval=mean_model2$coefficients[2,4],
                                   R2=mean_model2$adj.r.squared),
           Regr_Imputation=rbind(dbirwt_coef=reg_model2$coefficients[2],
                                   dbirwt_tval=reg_model2$coefficients[2,3],
                                   dbirwt_pval=reg_model2$coefficients[2,4],
                                   R2=reg_model2$adj.r.squared),
           Weighted_Model=rbind(dbirwt_coef=weight_model2$coefficients[2],
                                   dbirwt_tval=weight_model2$coefficients[2,3],
                                   dbirwt_pval=weight_model2$coefficients[2,4],
                                   R2=NA))
```

```
##             Complete_Case Mean_Imputation Regr_Imputation Weighted_Model
## dbirwt_coef   163.0750830     158.2618134     186.1990750      163.60420
## dbirwt_tval   102.4035643      94.3242156     124.6909909       97.31001
## dbirwt_pval     0.0000000       0.0000000       0.0000000        0.00000
## R2              0.3990157       0.3364693       0.4467038             NA
```

```
data.frame(C=cc_model2$coefficients[,1:2],
      M=mean_model2$coefficients[,1:2],
      R=reg_model2$coefficients[,1:2],
      W=weight_model2$coefficients[,1:2])
```

```
##               C.Estimate C.Std..Error    M.Estimate M.Std..Error   R.Estimate
## (Intercept) -3751.0862747   67.2229879 -3996.4261081   69.8012990 -4503.763462
## dgestat       163.0750830    1.5924747   158.2618134    1.6778492   186.199075
## dmage          10.7983692    0.8141250     9.7602806    0.7879593    11.230966
## dfage          -1.4119377    0.7165662    -0.8676849    0.6924699    -1.208868
## fmaps          40.5471721    5.0367016    85.5552624    4.7223643    24.476692
## wtgain          5.5017570    0.2378945     5.7022875    0.2286206     5.210718
## nprevis        -0.1952202    0.8794265     2.1116846    0.8412317    -1.276154
## cigar         -10.8149771    0.8899412   -11.5505728    0.8190295   -10.583927
## drink           2.0136090    9.7089149     2.1418475    7.4494182     3.245268
##             R.Std..Error    W.Estimate W.Std..Error
## (Intercept)   60.5074310 -3783.8104151   67.2673891
## dgestat        1.4932841   163.6041998    1.6812679
## dmage          0.7198434    10.9583187    0.8285288
## dfage          0.6323567    -1.3747842    0.7162890
## fmaps          4.3990078    40.6378238    5.4390939
## wtgain         0.2088858     5.4822402    0.2523684
## nprevis        0.7697021    -0.1246393    0.9236860
## cigar          0.7480217   -10.5325275    0.9187351
## drink          6.8025247     2.4994527    6.1913875
```

```
# NOTE: The regression models used *all* of the covariates in the data, the R^2
# values reflect that.
# NOTE: C stands for "Complete-Case", M for "Mean Imputation", R for "Regression
# Imputation", and W for "Weighted Model" for the second table
rm(cc_model2,mean_model2,reg_model2,weight_model2,WT_design,full_data)
```

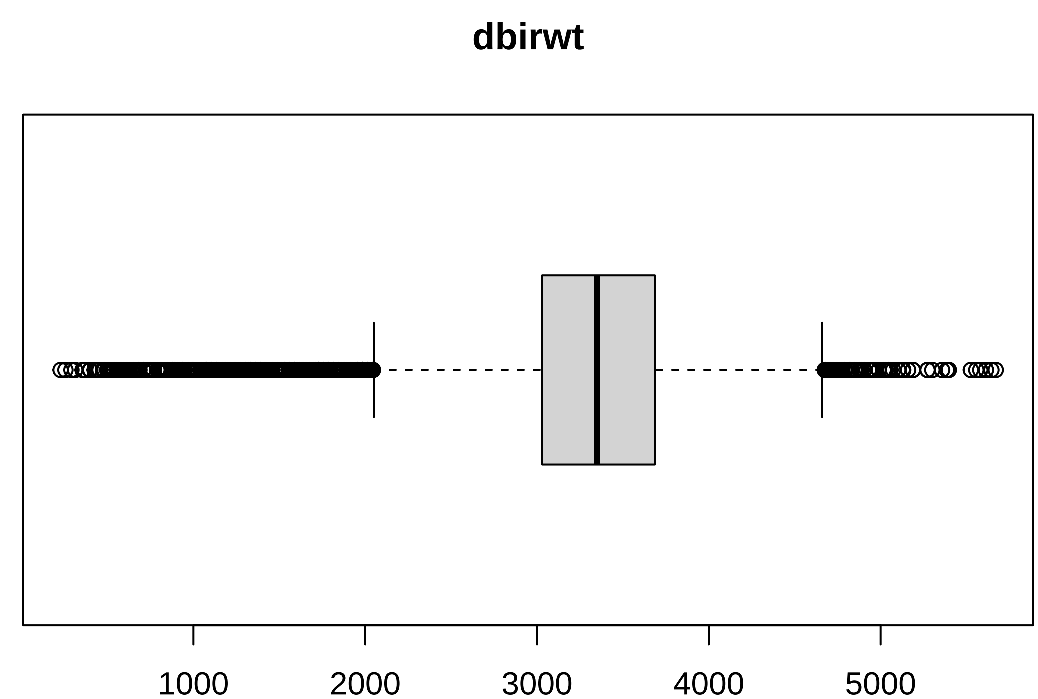
These results have a very similar interpretation to the previous problem. Mean imputation causes the attenuation of parameter estimates, though all approaches have the same conclusion - as **dgestat increases,**

dbirwt increases. Interestingly, nprevis is not significant at all in the complete case model, but is significant at level 0.05 for mean imputation and at level 0.1 for regression imputation. The covariate drink is insignificant for all three approaches, while all the other covariates are *very* significant in explaining the variance in dbirwt aside from dfage under mean imputation.
Once again, I would suggest complete case analysis or regression imputation *with the inclusion of random noise*. This is reasonable because the results of the regression are very similar across all three methods, meaning the missing values likely do not have a significant effect on the relationship between dbirwt and dgestat (at least using the methods we have covered so far). The counterargument paragraph from (4) also applies here - the preferred method is largely dependent on the exact research intentions of the user.
One last note is that these two regression results in tandem speak to the correlation between dgestat and dbirwt (as hinted at with the EM algorithm for mean estimation), since both are significant in explaining the other.

4.

The investigator is also interested in knowing the 25%, 50%, and 75%-tiles of the baby's birthweight (dbirwt) and their standard errors. How would you estimate them? As we covered in Lecture 5. This can be approached using bootstrap (10 pt).

```r
### Base Results
base_quant <- quantile(dge_data$dbirwt)
boxplot(dge_data$dbirwt,horizontal=T,main="dbirwt")
```



```r
# This was seen in the boxplots produced earlier, but it is clear that there are
# *many* outliers for this variable - indicated by the circles in the plot above.
# The above analysis does not include any standard error estimates, though, but
# this shortcoming can be addressed via bootstrap which will do repeated
# sampling of the variable.

### Bootstrap Approach
set.seed(1942)
boots <- 10000
boot_estimates <- matrix(0,nrow=boots,ncol=5)
for (i in 1:boots){
  # re-sample dbirwt values, keep same sample size
  values <- sample(dge_data$dbirwt,size=length(dge_data$dbirwt),replace=T)
  boot_estimates[i,] <- quantile(values)
}
data.frame(Base_Quantiles=base_quant,
           Bootstrapped=colMeans(boot_estimates), # calculate means
           Bootstrapped_SE=apply(boot_estimates,2, # calculate standard errors
                                 function(quantile){
                                   sd(quantile)/sqrt(boots)
                                   })
           ) %>% t()
```

```
##                           0%          25%          50%          75%         100%
## Base_Quantiles   227.0000000 3.030000e+03 3.350000e+03 3.686000e+03 5670.0000000
## Bootstrapped     243.7017000 3.025226e+03 3.354116e+03 3.686038e+03 5653.5826000
## Bootstrapped_SE    0.2728538 9.667948e-02 9.170813e-02 1.190302e-02    0.3052847
```

```r
rm(base_quant,boots,boot_estimates,i,values,dge_data)
```

Bootstrapping results of course vary from run to run, provided a set seed is not specified. With the chosen seed, the 25th percentile is 3025.22, the 50th is 3354.11, and the 75th is 3686.03. Notably, the maximum and minimum (100th and 0th percentiles) are considerably decreased and increased, respectively, bringing them closer to the median. This is to be expected because of the outliers demonstrated by the boxplot - repeated sampling of the same group of numbers is bound to have much less instances of these extreme values. Although each number is re-sampled uniformly (which raises another question for this inference task - sampling in some weighted way), there are far more values around the center quartiles than there are near the extremum, which makes the selection of values near said extremum less likely. In other words, the "candidate" values are much more highly concentrated in the center than at either of the tails. It also follows that the extremum have much higher standard errors than the middle quantiles, because of the many outliers.